

WaterSec AI Agent

Model Benchmark & Cost Analysis Report

Generated: 2026-05-10 07:10 | Data: Real IoT sensor data (158343 rows)

1 Groq Key Status

Key	Status
KEY_1	In cooldown
KEY_2	Available
KEY_3	Available
KEY_4	Available

Keys rotate automatically — rate-limited keys get a 60s cooldown before retry.

2 Overview

Compares 6 LLM candidates on 14 water domain prompts. **Real sensor data is injected into each prompt** so models answer with actual numbers. Scoring: $S = 0.25 \cdot R + 0.30 \cdot C + 0.25 \cdot I + 0.20 \cdot D - 0.5 \cdot H$ (Relevance, Correctness, Insight, Domain 1–5; Hallucination 0–2).

3 Models Tested

Model	Type	Speed	In \$/1M	Out \$/1M	Monthly \$
Llama 3.1 8B (Groq)	API	560 t/s	\$0.050	\$0.080	\$0.15
GPT-OSS 20B (Groq)	API	1000 t/s	\$0.100	\$0.500	\$0.66
Qwen3 32B (Groq)	API	535 t/s	\$0.290	\$0.590	\$1.01
Llama 3.3 70B (Groq)	API	280 t/s	\$0.590	\$0.790	\$1.63
Mistral 7B (Local)	Local	CPU	\$0.000	\$0.000	\$0.00
Qwen2.5 3B (Local)	Local	CPU	\$0.000	\$0.000	\$0.00

Monthly: 1,500 queries, avg 900 in + 700 out tokens. Hardware amortised 36mo/\$3,000.

4 Prompt-by-Prompt Results [real sensor data injected]

Advanced Prompts

[A1] Hard

Analyse the behavioural usage sequences detected in the residential CustomerC dataset within 5-minute windows. What do these sequences reveal about occupant behaviour and bathroom infrastructure? Provide specific water conservation recommendations backed by the occurrence numbers in the data.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.0s

Based on the data provided, we can analyze the behavioural usage sequences in residential CustomerC's dataset within 5-minute windows. **Key Observations:** 1. **High frequency of Sink usage:** The most common sequence is 'Sink → Sink' (30434 times), indicating that sinks are frequently used in the household. This could be due to various activities such as washing hands, brushing teeth, or food preparation. 2. **Tap usage often precedes Sink:** The second most common sequence is 'Tap → Sink' (25821 times), suggesting that taps are often used to fill sinks for various purposes. 3. **Flush usage is common:** The third most common sequence is 'Tap → Flush' (24428 times), indicating that toilets are frequently used in the household. 4. **Sink and Flush usage often follow each other:** Sequences like 'Sink → Flush' (16214 times) and 'Flush → Sink' (17732 times) suggest that occupants often use sinks and toilets in close proximity, possibly due to hygiene practices or simultaneous activities. 5. **Flush usage is often repeated:** The sequence 'Flush → Flush' (13160 times) indicates that toilets are frequently used in quick succession, which could be due to multiple users or a leaky toilet. **Occupant Behaviour Insights:** 1. **High water usage during peak hours:** The frequent usage of sinks and toilets during 5-minute windows suggests that water usage peaks during these periods. 2. **Occupants may not be aware of water conservation:** The high frequency of sequences like 'Sink...' [truncated]

GPT-OSS 20B (Groq) · 1.0s

Qwen3 32B (Groq) · 3.4s

<think> Okay, let's tackle this query. The user wants an analysis of the behavioral usage sequences from the CustomerC dataset, focusing on occupant behavior and bathroom infrastructure. They also need specific conservation recommendations based on the data. First, I need to look at the provided data table. The most frequent sequence is Sink → Sink with 30,434 occurrences. That's almost 19% of the total. Then Tap → Sink is next at 16.1%, and Tap → Flush at 15.2%. The least frequent is Tap → Tap at 6%. Hmm, Sink → Sink being the most common might indicate that people are using the sink multiple times in quick succession. Maybe they're washing hands, brushing teeth, or doing dishes. But why so often? If it's for brushing teeth, that could be a conservation opportunity because leaving the tap running while brushing uses a lot of water. The next thing is the high occurrence of Tap → Sink and Tap → Flush. Tap → Sink could be people filling a container at the sink after using the tap, but maybe they're not turning off the tap properly. Tap → Flush might be related to using the sink before flushing the toilet, but that doesn't make immediate sense. Wait, maybe it's about the sequence of events. If someone uses the tap and then the sink, perhaps they're washing hands after using the toilet, leading to the flush. Or maybe they're using the sink for something else before flushing. The Sink → Flush sequence is also notable. If someone uses the sink and then flushes, maybe they're... [truncated]

Llama 3.3 70B (Groq) · 1.4s

Based on the provided data, the most common sequence is 'Sink → Sink' with 30434 occurrences, accounting for 19% of the total frequency. This suggests that occupants often use the sink multiple times in a short period, possibly for activities like washing hands, brushing teeth, or rinsing face. The 'Tap → Sink' sequence, occurring 25821 times, implies that occupants frequently use the tap and then the sink, which could be for tasks like filling a glass or washing hands. The 'Tap → Flush' sequence, with 24428 occurrences, indicates that occupants often use the tap and then the toilet, possibly for personal hygiene. The presence of 'Flush → Sink' and 'Sink → Flush' sequences, with 17732 and 16214 occurrences respectively, suggests that occupants often use the toilet and then the sink, or vice versa, which is a common behaviour pattern in bathrooms. To conserve water, the following recommendations can be made: - Install low-flow faucets or aerators to reduce water flow rate, as the 'Sink → Sink' sequence is the most common, and reducing water flow can significantly decrease overall water consumption. - Consider installing a dual-flush toilet to reduce water usage per flush, as the 'Flush → Flush' sequence occurs 13160 times, indicating frequent toilet use. - Promote occupant behaviour change by encouraging them to turn off the tap while brushing teeth or washing hands, as the 'Tap → Sink' and 'Sink → Sink' sequences are common. - Install a faucet with a built-in timer or a s... [truncated]

Mistral 7B (Local) · 125.4s

Based on the analysis of the behavioral usage sequences detected in the CustomerC dataset within 5-minute windows, several patterns and insights can be drawn regarding occupant behavior and bathroom infrastructure. 1. The most frequent sequence is 'Sink → Sink' (30434 occurrences), which suggests that washing hands or brushing teeth is a common activity among the residents. This sequence also indicates that the sink is likely used more frequently than other fixtures in the bathroom. 2. The second and third most frequent sequences are 'Tap → Sink' (25821 occurrences) and 'Tap → Flush' (24428 occurrences), which may indicate that residents tend to run the tap before using it, possibly to wait for the water to warm up or to rinse something. This behavior could potentially be reduced by installing aerators on taps to save water. 3. The fourth most frequent sequence is 'Flush → Sink' (17732 occurrences), which may suggest that residents wash their hands after flushing the toilet. However, it's also possible that this sequence represents instances where the tap was left running while the toilet was being flushed, and then used for washing hands afterwards. In either case, this behavior could be reduced by encouraging residents to turn off the tap while not in use. 4. The fifth most frequent sequence is 'Sink → Flush' (16214 occurrences), which may indicate that residents flush the toilet after washing their hands or brushing their teeth. This sequence also suggests that the sin... [truncated]

Qwen2.5 3B (Local) · 177.0s

Analysis of Behavioural Usage Sequences The provided data reveals several insights about occupant behavior and bathroom infrastructure: 1. **Frequency of Events**: The most frequent sequence is "Sink → Sink" with 30,434 occurrences within a 5-minute window after the preceding event. This indicates that occupants frequently use the sink multiple times in quick succession. 2. **Common Usage Patterns**: - **Tap → Sink (25,821)**: Occupants often start by turning on the tap to rinse their hands or face before using the sink. - **Flush → Sink (17,732)**: After flushing the toilet, occupants frequently use the sink for hand washing and other hygiene purposes. - **Sink → Flush (16,214)**: Occupants often wash their hands after using the sink and then flush the toilet. ### Recommendations for Water Conservation Based on these sequences, here are specific water conservation recommendations: ##### 1. **Reduce Sink Usage** - **Install Dual-Flush Toilets**: These can be set to use less water for less urgent needs (e.g., a half-flush for urine and a full flush for solid waste). This reduces the frequency of Flush → Sink sequences. - **Implement Time-Outs on Faucets**: Install devices that automatically turn off taps after a certain period. For example, if occupants are using the sink continuously without turning it off, this could be set to shut off after 30 seconds. ##### 2. **Improve Toilet Efficiency** - **Dual-Flush Toilets**: These can significantly reduce water usage... [truncated]

[A2] Hard

Design a demand-based hot water heating schedule for the gym based on the real hourly sensor data. Identify peak hours, off-peak windows, and recommend specific pre-heating start times. Estimate expected energy savings as a percentage. Account for the fact that hot water usage is ~4x higher than col...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.2s

Based on the provided data, I will analyze the hot water usage patterns and design a demand-based hot water heating schedule for the gym. **Peak Hours Analysis**: The top 5 peak hours for hot water usage are: 1. 21:00 - 18,362.1 L 2. 20:00 - 15,040.7 L 3. 14:00 - 9,755.0 L 4. 09:00 - 9,482.0 L 5. 19:00 - 9,466.2 L **Off-Peak Windows**: To identify off-peak windows, I will look for hours with significantly lower hot water usage. Based on the data, the following hours have relatively low hot water usage: - 02:00 - 03:00 (average usage: 1,500 L) - 04:00 - 05:00 (average usage: 1,800 L) - 06:00 - 07:00 (average usage: 2,500 L) - 08:00 - 10:00 (average usage: 3,500 L, but with a peak at 09:00) **Pre-Heating Start Times**: To optimize energy savings, I recommend pre-heating the hot water system during off-peak windows. Based on the data, I suggest the following pre-heating start times: - 02:00 - 03:00: Pre-heat the system to maintain a moderate temperature (around 40°C) to ensure a quick response during peak hours. - 04:00 - 05:00: Gradually increase the temperature to around 50°C to meet the expected peak demand. - 06:00 - 07:00: Maintain the temperature at around 50°C to ensure a stable supply during the morning rush. - 08:00 - 10:00: Increase the temperature to around 60°C to meet the expected peak demand at 09:00. **Energy Savings Estimation**: To estimate the expected energy savings, I will assume that the hot water system is currently operating at full capacity during peak hours... [truncated]

GPT-OSS 20B (Groq) · 1.1s**Qwen3 32B (Groq) · 1.8s**

<think> Okay, let's tackle this problem step by step. The user wants a demand-based hot water heating schedule for a gym based on sensor data. They also need peak hours identified, off-peak windows, pre-heating start times, and energy savings estimates. Plus, they mentioned that hot water usage is four times higher than cold, so I need to account for that. First, I'll look at the provided data. The top 5 peak hours for hot water are 21:00, 20:00, 14:00, 09:00, and 19:00. The volumes are highest at 21:00 and 20:00, which are evening hours. The next highest is 14:00 (2 PM), then 09:00 and 19:00. So the gym seems to have high usage in the late afternoon and evening, plus a midday and morning peak. The flow rate data shows a median of 3.95 L/min, with a range from 0.51 to 15.06 L/min. That's a wide range, but the median is a good average to use for calculations. Now, the user wants a heating schedule. Since hot water usage is four times higher than cold, I need to adjust the schedule to prioritize heating during peak hours. But how does that affect the schedule? Maybe the heating system needs to be on during the peak hours to ensure hot water is available, and off during off-peak times to save energy. Peak hours are the top 5 listed. But the user might consider the top two as the main peaks. Let me check the data again. The first two hours (21:00 and 20:00) have significantly higher volumes than the others. So maybe those are the main peaks. The next three (14:00, 09:00, 19:00...) [truncated]

Llama 3.3 70B (Groq) · 1.8s

To design a demand-based hot water heating schedule, we need to analyze the peak hours and off-peak windows. Based on the provided data, the top 5 peak hours are 21:00, 20:00, 14:00, 09:00, and 19:00. The peak hours can be grouped into two main periods: - Morning peak: 09:00 - Evening peak: 19:00, 20:00, and 21:00 The off-peak windows are the periods between and outside these peak hours. To determine the pre-heating start times, we need to consider the time it takes to heat the water. However, the exact heating time is not provided in the data. Assuming the heating system can heat the water at a reasonable rate, we can estimate the pre-heating start times as follows: - Morning peak: pre-heat start time could be around 07:00-08:00 to ensure hot water is available by 09:00. - Evening peak: pre-heat start time could be around 16:00-17:00 to ensure hot water is available by 19:00. Since hot water usage is ~4x higher than cold water usage, the energy consumption for hot water is significantly higher. By implementing a demand-based hot water heating schedule, we can reduce energy consumption during off-peak windows. To estimate the expected energy savings, we need to consider the reduction in energy consumption during off-peak windows. Assuming the heating system is only active during peak hours and pre-heating periods, we can estimate the energy savings. Let's assume the heating system is active for 8 hours (2 hours of pre-heating + 6 hours of peak usage) and inactive f... [truncated]

Mistral 7B (Local) · 107.9s

To design a demand-based hot water heating schedule for the gym, we will first identify the peak hours and off-peak windows based on the provided data. Then, we will recommend specific pre-heating start times to optimize energy savings. Peak Hours: 1. 21:00 - 18,362.1 L (highest consumption) 2. 20:00 - 15,040.7 L 3. 14:00 - 9,755.0 L 4. 09:00 - 9,482.0 L 5. 19:00 - 9,466.2 L Off-Peak Windows: The remaining hours not included in the peak hours can be considered off-peak windows. However, since there is no data for some hours, it's essential to consider the flow rate distribution to ensure even coverage of off-peak hours. Pre-Heating Start Times: To account for the flow rate distribution and minimize cold water usage, we will pre-heat hot water during the following periods: 1. 18:00 - 23:59 (covers peak hours and some off-peak windows) 2. 06:00 - 08:59 (covers the 09:00 peak hour) 3. 13:00 - 14:59 (covers part of the 14:00 peak hour and some off-peak windows) Expected Energy Savings: Since hot water usage is approximately four times higher than cold water usage, we can assume that pre-heating during off-peak hours will significantly reduce energy consumption. However, to provide a more precise estimate, we would need data on the energy required for heating both hot and cold water. For now, let's conservatively estimate a 20% reduction in energy consumption by pre-heating during off-peak hours: Energy Savings = 20% * (Total Energy Consumption - Pre-Heated Energy Consumption... [truncated])

Qwen2.5 3B (Local) · 72.2s

To design a demand-based hot water heating schedule for the gym based on the provided data, we'll first identify peak hours and off-peak windows. Then, we will recommend specific pre-heating start times to minimize energy usage during peak hours while maintaining adequate hot water supply during off-peak hours. #### Step 1: Identify Peak Hours Based on the top 5 peak hours from the sensor data: - **21:00**: 18,362.1 L - **20:00**: 15,040.7 L - **14:00**: 9,755.0 L - **09:00**: 9,482.0 L - **19:00**: 9,466.2 L #### Step 2: Calculate Expected Energy Savings Given that hot water usage is approximately 4x higher than cold water usage: - If the gym uses 15,040.7 liters at 20:00 (peak), and assuming this is all hot water, then the total water used would be $\sqrt[4]{15,040.7} = 3,760.18$ liters. - The flow rate during peak hours can range from 0.51 to 15.06 L/min. #### Step 3: Estimate Energy Usage Assuming the median flow rate of 3.95 L/min: $\sqrt[4]{3,760.18} \times 3.95 = 946.63$ minutes (approximately 15.78 hours) - The pre-heating period would be the time before this usage occurs. #### Step 4: Recommend Pre-Heating Start Times To minimize energy usage during peak hours, we should start heating earlier: - **20:00** is a peak hour with 3,760.18 liters of hot water use. - If we assume the pre-heating period needs to be at least as long as this usage time (to ensure adequate supply), then the gym should start pre-heating by approximately 15.78 hours before the peak ho... [truncated]

[A3] Hard

CustomerB has a single aggregated sensor covering 2 WCs (man/woman), each with a central sink, 2 flushes, 2 flexibles, and 1 wudu sink — 9+ fixtures on 1 sensor. What are the specific risks of this architecture for leak detection and anomaly isolation? Redesign the sensor layout: state the minimum n...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.9s

Based on the provided data, I'll analyze the risks associated with the current architecture and propose a redesign for leak detection and anomaly isolation. **Current Architecture Risks:** 1. **False Positives:** With 9+ fixtures on a single sensor, it's challenging to isolate anomalies, leading to potential false positives. For example, a leak in one fixture might trigger an alert for the entire sensor. 2. **Insufficient Data:** A single sensor can't provide detailed information about each fixture's water usage, making it difficult to identify the source of anomalies. 3. **Limited Spatial Resolution:** A single sensor can't pinpoint the exact location of a leak or anomaly, requiring manual investigation and potentially delaying repairs. **Redesign Proposal:** To mitigate these risks, I recommend the following sensor layout: 1. **Minimum Number of Sensors:** 5 2. **Sensor Placement:** a. **Sensor 1:** Central sink (1 sensor) b. **Sensor 2:** Man WC (1 sensor) c. **Sensor 3:** Woman WC (1 sensor) d. **Sensor 4:** 2 Flushes + 2 Flexibles (1 sensor) e. **Sensor 5:** Wudu sink (1 sensor) This layout provides: **Improved Spatial Resolution:** Each sensor is dedicated to a specific area, allowing for more accurate anomaly detection and isolation. **Enhanced Data Collection:** With multiple sensors, you'll gather more detailed information about each fixture's water usage, making it easier to identify the source of anomalies. **Reduced False Positives:** By isolating ea... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem. The user is asking about the risks of having a single sensor for multiple fixtures in CustomerB's setup and how to redesign it. First, I need to understand the current setup. CustomerB has one sensor covering 9+ fixtures: 2 WCs (each with a central sink, 2 flushes, 2 flexibles), and 1 wudu sink. That's a lot of fixtures on a single sensor. The main risks here would be related to the sensor's ability to detect leaks and isolate anomalies. Since all fixtures are on one sensor, any leak in one area would be reported as a general issue without knowing the exact source. For example, if the wudu sink has a leak, the sensor can't tell it apart from a leak in the WC sinks. This makes troubleshooting difficult and time-consuming. Also, the aggregated data might mask smaller leaks. If multiple fixtures have minor leaks, the total usage might still be within normal ranges, making it hard to detect individual issues. Additionally, different fixtures have different usage patterns. The WC flushes might have high flow rates, while the wudu sink might have a steady flow. A single sensor might not capture these variations accurately, leading to false positives or negatives. Now, for the redesign. The goal is to minimize the number of sensors while improving detection and isolation. Let's break down the fixtures: - 2 WCs: Each WC has a central sink, 2 flushes, 2 flexibles. That's 5 fixtures per WC. So two WCs would be 10 fixtures. But the user me... [truncated]

Llama 3.3 70B (Groq) · 1.6s

The current architecture of CustomerB's water monitoring system, with a single aggregated sensor covering 9+ fixtures, poses several risks for leak detection and anomaly isolation. 1. **Leak detection difficulty**: With multiple fixtures connected to a single sensor, it will be challenging to identify the exact location of a leak. If a leak occurs, the sensor will detect an increase in water usage, but it will be difficult to determine which fixture is causing the leak. 2. **Anomaly isolation**: Similarly, if an anomaly occurs, such as a sudden increase in water usage, it will be hard to isolate the specific fixture or area causing the anomaly. 3. **False positives**: A single sensor covering multiple fixtures can lead to false positives, where a legitimate increase in water usage in one area is misinterpreted as a leak or anomaly in another area. 4. **Lack of granular data**: With a single sensor, the data collected will be aggregated, making it difficult to analyze water usage patterns for individual fixtures or areas. To redesign the sensor layout and mitigate these risks, I recommend a minimum of 4 sensors for CustomerB's setup: 1. **Sensor 1**: Place a sensor on the supply line to the 2 WCs, covering the 2 central sinks, 2 flushes, and 2 flexibles. This will allow for monitoring of water usage for the WCs and associated sinks. 2. **Sensor 2**: Place a sensor on the supply line to the wudu sink. This will provide dedicated monitoring for the wudu sink, allowing for ea... [truncated]

Mistral 7B (Local) · 103.7s

Given the current architecture of CustomerB's sensor setup, there are several potential risks for leak detection and anomaly isolation: 1. **Single point of failure**: Since there is only one sensor covering all fixtures, a malfunction or damage to this sensor could result in a loss of data for the entire system. This makes it difficult to detect leaks or anomalies promptly. 2. **Inadequate granularity**: With 9+ fixtures on a single sensor, it might be challenging to isolate specific anomalies or leaks to individual fixtures due to the lack of spatial resolution. For example, if there is a leak in one WC, it would be difficult to determine which WC it is without additional sensors. 3. **Limited capacity**: The high number of fixtures on a single sensor might exceed its capacity to accurately measure and monitor water consumption for each fixture individually. This could lead to inaccurate data and delayed detection of leaks or anomalies. To address these issues, I recommend redesigning the sensor layout as follows: 1. **Number of sensors needed**: At least 3 sensors would be necessary to cover all fixtures effectively. One sensor for each WC (man/woman), one for the central sink, and one for the wudu sink. This setup ensures that each fixture has its own dedicated sensor, reducing the risk of a single point of failure and improving the granularity of data collected. 2. **Placement of sensors**: Place one sensor under each WC (man/woman) to monitor water consumption for... [truncated]

Qwen2.5 3B (Local) · 49.3s

Analysis of CustomerB's Sensor Architecture ### Risks: **Leak Detection**: CustomerB has a single sensor covering all fixtures, which can lead to significant challenges in accurately detecting leaks. If one fixture is leaking, the entire system will show an increase in water usage that could be mistaken for normal activity or other issues. **Anomaly Isolation**: Isolating anomalies (e.g., unusual spikes) becomes difficult because a single sensor cannot differentiate between different fixtures. For example, if only one of the 9+ fixtures is leaking, it might not be immediately clear which fixture is causing the issue without additional data from multiple sensors. ### Redesigning Sensor Layout ### Minimum Number of Sensors Needed: To improve leak detection and anomaly isolation, we need to place sensors in a way that each fixture can be monitored individually. Given CustomerB has 9+ fixtures (2 WCs + 1 central sink + 2 flushes + 2 flexibles + 1 wudu sink), the minimum number of sensors needed is **9**. ### Sensor Placement: - **WCs**: Place a sensor in each WC to monitor water usage for both man and woman. - **Central Sink**: Place one sensor at the central sink. - **Flushes**: Place one sensor on each flush fixture. - **Flexibles**: Place one sensor on each flexible (assuming they are separate fixtures). - **Wudu Sink**: Place one sensor on the wudu sink. ### Detailed Sensor Placement: 1. **WC1** - Man 2. **WC1** - Woman 3. **Central Sink** 4. **Flush 1** 5. **Fl...** [truncated]

[A4] Expert

Rank the 4 monitored sites by leak detection risk (highest to lowest): gym showers (8 sensors, hot+cold), office toilets (4 aggregated bloc sensors), sanitary bloc (1 aggregated sensor), residential (7 granular sensors). For each site: (1) justify the ranking, (2) set an anomaly threshold in litres ...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.1s

Based on the provided data, I'll make some assumptions about the average session duration and consumption for each site. Since the actual session analysis failed, I'll use the provided average consumption values to estimate the leak detection risk. **Gym Showers (8 sensors, hot+cold)** 1. Justification: With 8 sensors monitoring hot and cold water, the gym showers have the highest risk of leak detection due to the increased number of sensors and potential for simultaneous leaks. 2. Anomaly threshold: 3.5 L (1.5 times the average consumption) 3. Alert action chain: - Initial alert: WaterSec AI sends a notification to the facilities manager within 5 minutes via email and SMS. - Follow-up alert: If the anomaly persists for 30 minutes, the facilities manager is notified via phone call. - Critical alert: If the anomaly exceeds 1 hour, the facilities manager and the building owner are notified via email and SMS. **Office Toilets (4 aggregated bloc sensors)** 1. Justification: With 4 aggregated bloc sensors, the office toilets have a moderate risk of leak detection due to the potential for simultaneous leaks. 2. Anomaly threshold: 2.5 L (1.1 times the average consumption) 3. Alert action chain: - Initial alert: WaterSec AI sends a notification to the facilities manager within 10 minutes via email and SMS. - Follow-up alert: If the anomaly persists for 45 minutes, the facilities manager is notified via phone call. - Critical alert: If the anomaly exceeds 1.5 hours, the faci... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Qwen3 32B (Groq) · 2.3s

<think> Okay, let's tackle this problem step by step. The user wants me to rank four sites by their leak detection risk from highest to lowest. The sites are gym showers, office toilets, sanitary bloc, and residential. Each has different numbers and types of sensors. First,

I need to understand what factors contribute to leak detection risk. More sensors might mean more data points, but if they're aggregated, it could be harder to detect small leaks. Granular sensors can pinpoint issues better. The average usage per session is also important—higher usage might mean more variability, making it harder to detect anomalies. Starting with the gym showers: they have 8 sensors (hot and cold), and the average session is 6.11 L. Since there are multiple sensors, maybe individual leaks can be detected more easily. But if a leak occurs in one of the 8, the system might catch it. However, the high average usage could mean that a leak might not be as obvious compared to lower usage areas. Office toilets have 4 aggregated bloc sensors. Aggregated means the data is combined, so a leak in one of the four might not be detected as quickly. The average customer session is 2.19 L. Lower usage might make anomalies stand out more, but since the sensors are aggregated, a leak in one of the four could be masked by normal usage in others. That's a risk because the aggregated data might not show a problem until it's significant. Sanitary bloc has 1 aggregated sensor. Only one sensor means any lea... [truncated]

Llama 3.3 70B (Groq) · 1.9s

To rank the monitored sites by leak detection risk, we need to consider the number of sensors, type of sensors, and the average water consumption per event at each site. 1. Residential (7 granular sensors): This site has the highest leak detection risk. With 7 granular sensors, it is likely that each sensor monitors a specific water outlet, providing detailed information about water usage. The high number of sensors increases the likelihood of detecting leaks, as each sensor can report anomalies in water consumption. - Anomaly threshold: 15 L per event (approximately 2-3 times the average consumption, assuming an average shower or tap usage) - Alert action chain: If an anomaly is detected, the building manager will be notified within 30 minutes via email and SMS. If the issue is not resolved within 2 hours, the maintenance team will be notified via phone call. 2. Gym showers (8 sensors, hot+cold): This site has the second-highest leak detection risk. The presence of 8 sensors monitoring both hot and cold water supplies provides comprehensive coverage of water usage. - Anomaly threshold: 20 L per event (approximately 3-4 times the average gym session consumption) - Alert action chain: If an anomaly is detected, the gym manager will be notified within 30 minutes via email and SMS. If the issue is not resolved within 2 hours, the maintenance team will be notified via phone call. 3. Office toilets (4 aggregated bloc sensors): This site has a moderate leak detect... [truncated]

Mistral 7B (Local) · 123.2s

Based on the data provided, let's rank the sites by leak detection risk from highest to lowest: 1. Gym Showers (8 sensors, hot+cold): The gym showers have the most sensors and a higher average consumption per event compared to other sites. This increased number of sensors allows for more precise monitoring, but also increases the likelihood of anomalies due to the higher water usage. Anomaly Threshold: 10 litres per event (10 L) Full Alert Action Chain: - Notify Site Manager via email and SMS immediately upon detection of an anomaly. - Send a follow-up email to the Maintenance Team with the details of the detected leak, including the sensor ID and timestamp. - If the leak persists for more than 30 minutes, escalate the alert to the Facilities Director via phone call. 2. Office Toilets (4 aggregated bloc sensors): The office toilets have a moderate average consumption per event and are monitored using aggregated bloc sensors. While these sensors may not provide as precise data as granular sensors, they still allow for effective leak detection due to the higher water usage compared to other sites. Anomaly Threshold: 5 litres per event (5 L) Full Alert Action Chain: - Notify Facilities Manager via email and SMS immediately upon detection of an anomaly. - If the leak persists for more than 15 minutes, escalate the alert to the IT Department via phone call, as there may be potential water damage affecting office equipment. 3. Sanitary Bloc (1 aggreg... [truncated]

Qwen2.5 3B (Local) · 55.6s

Given the data provided and the lack of "module 'tools' has no attribute 'session_analysis'", I will proceed with a manual analysis based on available information. ### Site Rankings by Leak Detection Risk (Highest to Lowest) 1. **Gym Showers**: - **Justification**: The gym showers have 8 sensors, which means they are likely monitoring multiple users simultaneously during each session. With an average session of 6.11 L and a high number of sensors, the potential for detecting leaks is significant. - **Anomaly Threshold**: Set at 20 L per event (to account for variations in usage). - **Alert Action Chain**: - Notify Maintenance Team: Within 30 minutes via email and SMS. - Conduct On-Site Inspection within 1 hour. 2. **Office Toilets**: - **Justification**: The office toilets have 4 aggregated bloc sensors, which means they are monitoring a group of users (likely multiple people using the same toilet at different times). With an average session of 2.19 L and fewer sensors, the risk is lower but still significant. - **Anomaly Threshold**: Set at 10 L per event. - **Alert Action Chain**: - Notify Maintenance Team: Within 30 minutes via email. - Conduct On-Site Inspection within 6 hours. 3. **Sanitary Bloc**: - **Justification**: The sanitary bloc has a single aggregated sensor, which means it is monitoring multiple users (likely the entire block). With an average session of 2.19 L and only one sensor, the risk is relatively low. - ... [truncated]

[A5] Expert

Residential Flush 2 recorded 2796.63 L in a single event (z-score: 11.75). A normal flush is 6 L. Flow rate during this event was 6 L/min. Calculate step by step: 1. Exact duration of the event in minutes and hours 2. Water cost at 0.003 EUR/L 3. CO2 equivalent at 0.298 kg CO2 per m³ of water...

Llama 3.1 8B (Groq) · 0.7s

To calculate the duration of the event, we need to know the flow rate and the total volume of water used. 1. Given flow rate = 6 L/min and total volume = 2796.63 L, we can calculate the duration in minutes: $2796.63 \text{ L} / 6 \text{ L/min} = 466.1$ minutes To convert minutes to hours, we divide by 60: $466.1 \text{ minutes} / 60 = 7.77$ hours 2. To calculate the water cost, we multiply the total volume by the cost per liter: $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.39$ EUR 3. To calculate the CO2 equivalent, we first need to convert the volume from liters to cubic meters: $2796.63 \text{ L} / 1000 = 2.79763 \text{ m}^3$ Then, we multiply the volume in cubic meters by the CO2 equivalent per cubic meter: $2.79763 \text{ m}^3 * 0.298 \text{ kg CO2/m}^3 = 0.834 \text{ kg CO2}$ Now, let's rank the three most likely root causes by probability: 1. **Blocked or clogged drain line** (probability: high) The extremely high volume of water used suggests a blockage or clog in the drain line, causing the water to flow back into the toilet bowl and continue flowing until the tank is empty. This would explain the prolonged duration of the event. 2. **Improperly installed or faulty toilet flapper** (probability: medium) A faulty or improperly installed toilet flapper can cause the toilet to continuously run, leading to excessive water usage. However, this would typically not result in such a high volume of water being used. 3. **Malfunctioning or stuck float valve** (probability: low) A malfunctioning or stuck float valve can cause the toilet to continuously run, but... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Step-by-step calculations | Item | Value | Units | |-----|-----|-----| | 1. Duration (minutes) | $2796.63 \div 6 = 466.105$ | min | | 1. Duration (hours) | $466.105 \div 60 = 7.768$ | h | | 2. Water cost | $2796.63 \times 0.003 = 8.38989$ | EUR | | 3. CO2 equivalent | $2796.63 \text{ L} = 2.79663 \text{ m}^3$; $2.79663 \times 0.298 = 0.8334$ | kg CO2 | --- ### Likely root cause ranking (most to least probable) | Rank | Root cause | Why it fits | Probability (qualitative) | |-----|-----|-----|-----| | 1 | **Faulty flush valve / continuous flow leak** | A single flush event produced 2796 L at the normal 6 L/min rate, meaning the water was running for almost 8 h. The only way a toilet can keep flowing that long is a mechanical failure (e.g., a broken flapper or valve that never closes). | Very high | | 2 | **User left the toilet open or repeatedly flushed** | If the user accidentally left the toilet open or performed many flushes in quick succession, the system would record a single long event. However, the event is logged as one continuous flow, making this less likely than a mechanical fault. | Moderate | | 3 | **Sensor / data logging error** | A mis-calibrated flow meter or a software glitch could report an inflated volume or duration. This is possible but would usually affect multiple events, not just one. | Low | --- ### WaterSec alert that should have fired - **High Water Consumption – Flush Event** *Trigger: Any single flush event > 100 L (... [truncated]

Qwen3 32B (Groq) · 1.7s

<think> Okay, let's tackle this problem step by step. First, I need to calculate the duration of the event. The user mentioned that the flow rate was 6 L/min, and the total consumption was 2796.63 L. To find the duration in minutes, I'll divide the total liters by the flow rate. So 2796.63 divided by 6 gives me the minutes. Then, converting that to hours by dividing by 60. Next, calculating the water cost. The cost is 0.003 EUR per liter. Multiplying the total liters by 0.003 will give the total cost in euros. For the CO2 equivalent, the rate is 0.298 kg CO2 per cubic meter. Since 1 cubic meter is 1000 liters, I need to convert 2796.63 liters to cubic meters by dividing by 1000. Then multiply by 0.298 to get the CO2 in kg. Now, ranking the root causes. The z-score is 11.75, which is way above normal. A normal flush is 6 L, but this event used over 2700 L. Possible causes could be a stuck valve, a broken toilet, or a pipe leak. A stuck valve would keep water flowing until manually stopped. A broken toilet might have a continuous leak. A pipe leak could be a slow drip but over time adds up. The most likely is a stuck valve because it's a sudden high usage. Then a broken toilet, and a pipe leak is less likely but possible. The WaterSec alert should have triggered a high usage alert due to the z-score. </think> 1. **Event Duration Calculation** - **Total consumption**: 2796.63 L - **Flow rate**: 6 L/min - **Duration in minutes**: $2796.63 / 6 = 466.105$ minutes ... [truncated]

Llama 3.3 70B (Groq) · 2.0s

To calculate the required values, we will follow the steps provided. 1. Exact duration of the event in minutes and hours: First, we need to calculate the duration of the event in minutes. We know that consumption_L = 2796.63 L and the flow rate = 6 L/min. To find the duration, we divide the total litres used by the flow rate: $2796.63 \text{ L} / 6 \text{ L/min} = 466.1$ minutes. To convert minutes to hours, we divide by 60: $466.1 \text{ minutes} / 60 = 7.7683$ hours. 2. Water cost at 0.003 EUR/L: To find the water cost, we multiply the total litres used by the cost per litre: $2796.63 \text{ L} * 0.003 \text{ EUR/L} = 8.38989$ EUR. 3. CO2 equivalent at 0.298 kg CO2 per m³ of water: First, we need to convert litres to cubic meters. Since 1 cubic meter = 1000 litres, we divide the total litres used by 1000: $2796.63 \text{ L} / 1000 = 2.79663 \text{ m}^3$. Then, we multiply the volume in cubic meters by the CO2 equivalent per cubic meter: $2.79663 \text{ m}^3 * 0.298 \text{ kg CO2/m}^3 = 0.8333 \text{ kg CO2}$. Now, let's rank the three most likely root causes by probability and explain each: 1. **Leak or open tap**: This is the most likely root cause with a high probability. The z-score of 11.75 indicates that the event is extremely unusual, and the large volume of water used (2796.63 L) suggests that the tap might have been left open or there could be a leak in the system. 2. **Incorrect flow rate sensor reading**: This is another possible root cause, but with a lower probability. If the flow rate sensor is malfunctioning, it could be reporting an inco... [truncated]

Mistral 7B (Local) · 85.0s

1. To find the duration of the event in minutes, we first need to convert litres to cubic meters (m³) because the flow rate is given in m³/min. Since 1 m³ equals 1000 L, the consumption_L of 2796.63 L is equal to 2.79663 m³. Now we can calculate the duration by dividing the total volume used (2.79663 m³) by the flow rate (6 L/min), which equals 0.4661 minutes. To convert this to hours, we divide by 60 (minutes per hour): $0.4661 / 60 = 0.007768$ hours or approximately 0.0078 hours. 2. To calculate the water cost for this event, multiply the consumption_L (2796.63 L) by the price per litre (0.003 EUR/L), which equals 8.38999 EUR. 3. The CO2 equivalent can be

calculated by multiplying the total volume used (2.79663 m³) by the CO2 emission factor (0.298 kg CO2 per m³ of water), which equals 0.8314 kg CO2. Ranking the three most likely root causes: 1. Excessive usage: The consumption_L is significantly higher than a normal flush (6 L). This suggests that there might be excessive water use, such as a long shower or running the tap for an extended period. WaterSec alert that should have fired: "Excessive water usage detected." 2. Leakage: A sudden high consumption_L could also indicate a leak in the plumbing system. While it's possible that this was a single excessive flush, it's worth investigating if there are any signs of ongoing leaks. WaterSec alert that should have fired: "Potential water leak detected." 3. Inefficient appliance: If the event involved using an appliance I... [truncated]

Qwen2.5 3B (Local) · 47.5s

1. **Exact duration of the event in minutes and hours:** - Given that the flow rate was 6 L/min and the total consumption was 2796.63 L: $\frac{\text{Duration (in minutes)}}{\text{Flow Rate}} = \frac{2796.63}{6} = 466.1 \text{ minutes}$ - Convert the duration to hours: $\frac{\text{Duration (in hours)}}{60} = \frac{466.1}{60} = 7.77 \text{ hours}$ 2. **Water cost at 0.003 EUR/L:** - Total consumption in litres is 2796.63 L. - Cost calculation: $\text{Cost (in EUR)} = 2796.63 \times 0.003 = 8.40 \text{ EUR}$ 3. **CO2 equivalent at 0.298 kg CO2 per m³ of water:** - Convert litres to cubic meters (1 L = 0.001 m³): $\text{Total Consumption in m}^3 = 2796.63 \times 0.001 = 2.79663 \text{ m}^3$ - CO2 calculation: $\text{CO}_2 \text{ Equivalent (in kg)} = 2.79663 \times 0.298 = 0.845 \text{ kg}$ 4. **Ranking the most likely root causes by probability:** 1. **High Flow Rate:** The event had a flow rate of 6 L/min, which is significantly higher than the normal flush (6 L). This suggests that there might have been an accidental or intentional high-flow situation such as using a hose to flush out pipes or fixtures. 2. **Water Supply System Issue:** There could be a malfunction in the water supply system leading to a prolonged flow event. For example, a leaky valve or a faulty shut-off mechanism could cause continuous water flow for extended periods. 3. **User ...** [truncated]

[A6] Expert

A water utility wants to use WaterSec to meet ISO 24512 water service management standards. Based on the 4 customer profiles and their sensor granularity, identify: (1) which ISO 24512 KPIs can be computed directly from this sensor data and how, (2) which KPIs require additional data sources — name ...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.1s

Based on the provided data, I can identify the ISO 24512 KPIs that can be computed directly from this sensor data and propose a monthly compliance report structure. **Directly Computable KPIs:** 1. **Water Consumption (KPI 1.1.1):** Total litres used by each customer over the given period. - customerA: 105,712.5 L - customerB: 214,731.3 L - customerC: 162,082.1 L - gym: 162,679.6 L 2. **Average Daily Water Consumption (KPI 1.1.2):** Total litres used by each customer divided by the number of days. - customerA: $\frac{105,712.5 \text{ L}}{175 \text{ days}} = 604.1 \text{ L/day}$ - customerB: $\frac{214,731.3 \text{ L}}{214 \text{ days}} = 1003.4 \text{ L/day}$ - customerC: $\frac{162,082.1 \text{ L}}{628 \text{ days}} = 258.1 \text{ L/day}$ - gym: $\frac{162,679.6 \text{ L}}{746 \text{ days}} = 218.1 \text{ L/day}$ 3. **Total Water Consumption (KPI 1.1.3):** Total litres used by all customers over the given period. - Total: $105,712.5 \text{ L} + 214,731.3 \text{ L} + 162,082.1 \text{ L} + 162,679.6 \text{ L} = 645,205.5 \text{ L}$ 4. **Average Daily Total Water Consumption (KPI 1.1.4):** Total litres used by all customers divided by the total number of days. - Total: $\frac{645,205.5 \text{ L}}{175 \text{ days} + 214 \text{ days} + 628 \text{ days} + 746 \text{ days}} = \frac{645,205.5 \text{ L}}{1763 \text{ days}} = 3.66 \text{ L/day}$ **KPIs Requiring Additional Data Sources:** 1. **Water Loss (KPI 2.1.1):** Requires data on the total water production and the total water consumption. - Additional data source: Water production data from the utility. 2. **Water Pressure (KPI 3.1.1):** Requires data on the water pressure at the customer's premises. - Additional data source: Pressure sensors ins... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem step by step. The user wants to use WaterSec's sensor data to meet ISO 24512 standards. First, I need to recall what ISO 24512 entails. From what I remember, ISO 24512 is about water service management, focusing on performance indicators like water supply reliability, efficiency, and customer satisfaction. The data provided includes four customers with daily consumption and total over different periods. The task is to identify which KPIs can be calculated directly from this data, which need additional sources, and then structure a monthly report. Starting with the first part: KPIs that can be computed directly. The sensor data gives daily usage and total over a period. So, average daily consumption per customer is straightforward. For example, customerA has 604.1 L/day. That's a direct KPI. Total water supplied per customer is also given, like 105,712.5 L for customerA over 175 days. Another KPI could be the average daily consumption per customer type. Since there are different profiles (customerA, B, C, gym), grouping them might help. Also, the total water supplied per customer type can be calculated by summing their totals. For instance, gyms have 162,679.6 L over 746 days. But wait, some KPIs might require more data. For example, water loss or non-revenue water isn't directly available here. The sensor data only tracks consumption, not losses. So that would need additional sources like metering data from distribution networks or... [truncated]

Llama 3.3 70B (Groq) · 2.2s

To meet ISO 24512 water service management standards, we will analyze the provided sensor data from WaterSec. **(1) ISO 24512 KPIs computable from sensor data:** - Average daily water consumption per customer: already provided in the data (e.g., customerA: 604.1 L/day). - Total water consumption per customer over a period: also provided in the data (e.g., customerA: 105,712.5 L over 175 days). - Water consumption per day for each customer profile: given (e.g., customerA: 604.1 L/day). - To calculate the average water consumption per customer per month, we need to know the number of days in the month. Assuming a month has approximately 30 days, we can estimate: - customerA: $604.1 \text{ L/day} \times 30 \text{ days} = 18,123 \text{ L/month}$ - customerB: $1003.4 \text{ L/day} \times 30 \text{ days} = 30,102 \text{ L/month}$ - customerC: $258.1 \text{ L/day} \times 30 \text{ days} = 7,743 \text{ L/month}$ - gym: $218.1 \text{ L/day} \times 30 \text{ days} = 6,543 \text{ L/month}$ **(2) KPIs requiring additional data**

sources:** - Water pressure and flow rate: require additional sensor data or hydraulic modeling. - Water quality parameters (e.g., pH, turbidity, bacteria): need water quality monitoring data from laboratories or in-situ sensors. - Customer satisfaction: requires survey data from customers. - Water service continuity (e.g., hours of supply per day): needs data on supply hours, outages, and duration from SCADA systems or customer reports. - Water losses (e.g., leakage rates): requires data from leakage detection surveys, night flow measurements, or DMA (District Metered Area)... [truncated]

Mistral 7B (Local) · 110.2s

(1) ISO 24512 KPIs that can be computed directly from the sensor data are: - Average Daily Consumption (ADC): Calculate the average consumption per day for each customer by dividing the total consumption by the number of days. - For example, ADC for CustomerA = 105712.5 L / 175 days = 604.1 L/day (already provided in the data) - Total Consumption (TC): This is already provided in the data for each customer. (2) ISO 24512 KPIs that require additional data sources are: - Peak Daily Consumption (PDC): To calculate PDC, you need to know the maximum consumption per day within a specified time period. This requires historical data with a higher sensor granularity than daily totals. - Consumption Percentiles (CP): Calculating percentiles requires more frequent and consistent data points throughout the day, which may not be available with the current sensor setup. (3) Monthly Compliance Report Structure: Title: Water Utility Monthly Compliance Report Sections: 1. Executive Summary: - Brief overview of key findings and compliance status for each KPI. - Highlight any significant changes or trends in water consumption across customers. 2. Customer Consumption Analysis: - Average Daily Consumption (ADC) for each customer. - Total Consumption (TC) for each customer. 3. Peak Daily Consumption (PDC) Analysis: - PDC calculation methodology and results for each customer (if available). - Recommendations for improving data granularity to accurately calculate PDC.... [truncated]

Qwen2.5 3B (Local) · 66.2s

(1) Directly Computable ISO 24512 KPIs ##### Customer Profiles: - **Customer A**: 604.1 L/day - **Customer B**: 1003.4 L/day - **Customer C**: 258.1 L/day - **Gym**: 218.1 L/day **KPIs that can be computed directly from sensor data:** 1. **Average Daily Consumption (ADC)**: - ADC = Total consumption / Number of days - For Customer A: ADC = 105,712.5 / 175 ≈ 604.1 L/day - For Customer B: ADC = 214,731.3 / 214 ≈ 1003.4 L/day - For Customer C: ADC = 162,082.1 / 628 ≈ 258.1 L/day - For Gym: ADC = 162,679.6 / 746 ≈ 218.1 L/day #### (2) KPIs Requiring Additional Data Sources ##### Customer Profiles: - **Customer A**: 604.1 L/day | Total: 105,712.5 L over 175 days - **Customer B**: 1003.4 L/day | Total: 214,731.3 L over 214 days - **Customer C**: 258.1 L/day | Total: 162,082.1 L over 628 days - **Gym**: 218.1 L/day | Total: 162,679.6 L over 746 days **KPIs that require additional data sources:** 1. **Water Usage Variability (UV)**: - UV = Standard Deviation of daily consumption - Requires historical data to calculate standard deviation. 2. **Peak Hour Consumption (PHC)**: - PHC = Maximum value in the hourly consumption data. - Requires hourly granularity data, which is not provided here. 3. **Water Efficiency Index (WEI)**: - WEI = ADC / UV - Requires UV to calculate. 4. **Total Water Usage Variability (TWUV)**: - TWUV = Standard Deviation of total consumption over the period. - Requires historical data for total consumption, which is ... [truncated]

Standard Prompts

[S1] Easy

What is the total water consumption at the gym last month? The gym has 4 shower cabins, each with a hot and cold sensor (8 sensors total). Analyse the monthly trend and state the most recent full month total in litres.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.5s

To find the total water consumption at the gym last month, we need to find the most recent full month total in litres. The most recent full month is 2026-04-30. The consumption for this month is 10898.8 L. To find the total water consumption for the gym, we need to multiply the consumption for the most recent month by the number of months. However, we do not have the consumption data for all months. We can only calculate the total consumption for the months where we have data. Here are the total consumption values for the months where we have data: - 2024: 1.28 + 0 + 0 + 849.47 = 850.75 L - 2025: 6555.26 + 2604.66 + 3323.2 + 3257 + 1792 + 7990.48 + 6440.59 + 3847.6 + 6100.08 + 5724.53 + 1311.83 + 3075.66 + 9697.42 + 9954.82 + 14118.6 + 12331.9 + 12136.4 = 104,570.11 L - 2026: 10898.8 + 7825.56 + 15387.7 + 14540.2 + 2914.66 = 51,066.72 L We can now calculate the total water consumption for the gym from 2024 to 2026: Total consumption = 850.75 + 104,570.11 + 51,066.72 = 156,487.58 L So, the total water consumption at the gym from 2024 to 2026 is approximately 156,487.58 L.

GPT-OSS 20B (Groq) · 1.0s

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's see. The user is asking for the total water consumption at the gym last month. They mentioned the gym has 4 shower cabins with 8 sensors total. The data provided is from WaterSec sensors, with periods from 2024-04-30 up to 2026-05-31. Each row has consumption_L and flow_rate_Lpm. First, I need to figure out what the most recent full month is. The last date in the data is 2026-05-31. Since the current date isn't specified, I'll assume that the latest month is May 2026. But wait, the user is asking for "last month," which would depend on when the current date is. However, since the data goes up to May 2026, and there's no data after that, I'll take May 2026 as the most recent full month. Looking at the data, the consumption_L for 2026-05-31 is 2914.66 liters. That should be the total for May. Now, I need to check if there's any other data that might be relevant. The user mentioned 4 shower cabins with 8 sensors, but the data provided seems to aggregate all sensors into a single monthly consumption. So each row's consumption_L is the total for all sensors that month. Therefore, the May 2026 entry is already the total for the gym's showers that month. I should also analyze the monthly trend. Looking at the consumption over the months, there's a general upward trend from April 2024 to January

2026, peaking at 15387.7 liters in January 2026. Then there's a drop in February 2026 to 14540.2 liters, followed by a decrease in March 2026 to 7825.56 liters. April 2... [truncated]

Llama 3.3 70B (Groq) · 0.6s

To find the total water consumption at the gym last month, we need to look at the most recent full month data. The most recent full month is 2026-04-30. The total water consumption for the month of April 2026 is 10898.8 litres. To analyze the monthly trend, we can look at the consumption for the previous months: - March 2026: 7825.56 litres - February 2026: 14540.2 litres - January 2026: 15387.7 litres - December 2025: 12136.4 litres - November 2025: 12331.9 litres - October 2025: 14118.6 litres The total water consumption has been fluctuating over the months, with the highest consumption in January 2026 at 15387.7 litres and the lowest in June 2025 at 1311.83 litres. The most recent full month total in litres is 10898.8 litres for April 2026.

Mistral 7B (Local) · 115.7s

To find the total water consumption at the gym last month (February 2025), we need to sum up the consumption for all the days in February. However, there is no data available for the last day of February (2025-02-28). Therefore, I can only provide the total water consumption from the beginning of February until the last available data point (2026-03-31). Here's the calculation: 1. Sum up the consumption for each day from 2025-02-01 to 2025-02-28 (excluding 2025-02-28 since there is no data available): - From 2025-02-01 to 2025-02-27, the total consumption is: 6440.59 + 3847.6 + 3323.2 + 3257 + 1792 + 7990.48 + 6440.59 = 31,133.53 L - Since there is no data for 2025-02-28, the total consumption for February remains as 31,133.53 L. So, the total water consumption at the gym last month (February 2025) was approximately 31,133.53 liters.

Qwen2.5 3B (Local) · 33.5s

To determine the total water consumption at the gym last month, we need to identify the most recent full month's data and sum up its consumption values. From the provided DATA, the most recent full month is April 2026 with a consumption of 10898.8 litres. Therefore, the total water consumption at the gym last month (April 2026) was **10898.8 litres**.

[S2] Easy

Compare hot vs cold water usage across all 4 gym shower cabins. Calculate the hot-to-cold ratio per cabin and identify which cabin uses the most hot water overall. What does this suggest operationally?

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.7s

To compare hot vs cold water usage across all 4 gym shower cabins, we'll calculate the hot-to-cold ratio per cabin and identify which cabin uses the most hot water overall. ****Hot-to-Cold Ratio per Cabin:**** Cabin 1: 4.3x (Hot: 45,386.4 L / Cold: 10,664.1 L) Cabin 2: 5.6x (Hot: 39,222.8 L / Cold: 7,039.8 L) Cabin 3: 4.0x (Hot: 30,778.5 L / Cold: 7,729.5 L) Cabin 4: 2.8x (Hot: 16,164.9 L / Cold: 5,693.5 L) ****Cabin with the Highest Hot Water Usage:**** To determine which cabin uses the most hot water overall, we'll compare the total hot water usage for each cabin. Cabin 1: 45,386.4 L Cabin 2: 39,222.8 L Cabin 3: 30,778.5 L Cabin 4: 16,164.9 L Cabin 1 uses the most hot water overall, with a total of 45,386.4 L. ****Operational Implications:**** The high hot water usage in Cabin 1 suggests that this cabin is being used more frequently or for longer periods, or that the showerheads in this cabin are more efficient at heating water. This could be due to various factors, such as: * Higher demand for hot showers in this cabin * More users in this cabin * More frequent or longer showers in this cabin * More efficient showerheads in this cabin To optimize water usage and reduce energy consumption, the gym management could consider: * Installing low-flow showerheads in all cabins * Implementing a timer or flow restrictor to limit shower duration * Encouraging users to take shorter showers * Monitoring water usage and adjusting the showerheads or water heaters as needed.

GPT-OSS 20B (Groq) · 1.0s

****Hot to Cold Water Ratios (per cabin)**** | Cabin | Hot water (L) | Cold water (L) | Ratio (Hot ÷ Cold) |
|-----|-----|-----|-----| | 1 | 45 386.4 | 10 664.1 | 4.26 ≈ 4.3x | | 2 | 39 222.8 | 7 039.8 | 5.57 ≈ 5.6x | | 3 | 30 778.5 | 7 729.5 | 3.98 ≈ 4.0x | | 4 | 16 164.9 | 5 693.5 | 2.84 ≈ 2.8x | ****Cabin with the highest total hot water use**** - Cabin 1 uses the most hot water overall: 45 386.4 L, which is higher than Cabin 2 (39 222.8 L), Cabin 3 (30 778.5 L), and Cabin 4 (16 164.9 L). ****Operational implications**** 1. ****Energy cost focus**** - Cabin 1's high hot water volume means it consumes the most energy for heating. Targeting this cabin for efficiency upgrades (e.g., low flow showerheads, better insulation, or a lower set point temperature) could yield the biggest savings. 2. ****User behaviour**** - The high hot to cold ratio in Cabin 2 (5.6x) suggests users there may be running showers longer or at higher temperatures. Providing signage or a brief user guide on efficient shower use could reduce hot water demand. 3. ****Maintenance check**** - Verify that thermostatic mixing valves and temperature controls are functioning correctly in all cabins, especially Cabin 1, to ensure that hot

Qwen3 32B (Groq) · 1.8s

<think> Okay, let's tackle this problem step by step. The user wants me to compare hot and cold water usage across four gym shower cabins, calculate the hot-to-cold ratio for each, and determine which cabin uses the most hot water overall. Then, I need to interpret what this suggests operationally. First, I'll look at the data provided. Each cabin has hot and cold water usage in liters, along with the ratio already calculated. Wait, the ratios are given, but maybe I should verify them to ensure accuracy. Let me check each cabin's ratio. Starting with Cabin 1: Hot is 45,386.4 L and Cold is 10,664.1 L. The ratio is Hot divided by Cold. So 45,386.4 / 10,664.1. Let me calculate that. Dividing both numbers by 1000 to simplify: 45.3864 / 10.6641 ≈ 4.257. The given ratio is 4.3x, which matches. Good. Cabin 2: Hot 39,222.8 / Cold 7,039.8. Let's compute that. 39,222.8 / 7,039.8 ≈ 5.573. The given ratio is 5.6x. That's correct. Cabin 3: 30,778.5 / 7,729.5 ≈ 4.0. Exactly 4.0x as given. Perfect. Cabin 4: 16,164.9 / 5,693.5 ≈ 2.838. The given ratio is 2.8x. Close enough. So the ratios provided are accurate. Now, the next part is to identify which cabin uses the most hot water overall. Looking at the hot water

volumes: Cabin 1: 45,386.4 L Cabin 2: 39,222.8 L Cabin 3: 30,778.5 L Cabin 4: 16,164.9 L Comparing these numbers, Cabin 1 has the highest hot water usage. So that's straightforward. Now, the operational implications. The user wants to know what this suggests. Let's think.... [truncated]

Llama 3.3 70B (Groq) · 1.0s

To compare hot vs cold water usage across all 4 gym shower cabins, we need to examine the provided data. Cabin 1 uses 45,386.4 L of hot water and 10,664.1 L of cold water, resulting in a hot-to-cold ratio of 4.3. Cabin 2 uses 39,222.8 L of hot water and 7,039.8 L of cold water, resulting in a hot-to-cold ratio of 5.6. Cabin 3 uses 30,778.5 L of hot water and 7,729.5 L of cold water, resulting in a hot-to-cold ratio of 4.0. Cabin 4 uses 16,164.9 L of hot water and 5,693.5 L of cold water, resulting in a hot-to-cold ratio of 2.8. The cabin with the highest hot water usage is Cabin 1, with 45,386.4 L. This suggests operationally that Cabin 1 may have a higher frequency of use or longer shower durations, and may benefit from water-saving measures such as low-flow showerheads or timers to reduce hot water consumption. In terms of the hot-to-cold ratio, Cabin 2 has the highest ratio of 5.6, indicating a strong preference for hot water in this cabin. Cabin 4 has the lowest ratio of 2.8, suggesting a more balanced use of hot and cold water. Overall, the data suggests that there are opportunities to optimize water usage in the gym shower cabins, particularly in Cabin 1, to reduce hot water consumption and improve overall water efficiency.

Mistral 7B (Local) · 90.8s

To compare hot vs cold water usage across all four gym shower cabins, we'll first calculate the hot-to-cold ratios for each cabin. Cabin 1: Hot-to-cold ratio = $45386.4 \text{ L} / 10664.1 \text{ L} = 4.27\times$ Cabin 2: Hot-to-cold ratio = $39222.8 \text{ L} / 7039.8 \text{ L} = 5.56\times$ Cabin 3: Hot-to-cold ratio = $30778.5 \text{ L} / 7729.5 \text{ L} = 4.0\times$ Cabin 4: Hot-to-cold ratio = $16164.9 \text{ L} / 5693.5 \text{ L} = 2.85\times$ Now, let's identify which cabin uses the most hot water overall by finding the maximum hot water usage among all cabins. Cabin 1: Hot water usage = 45386.4 L Cabin 2: Hot water usage = 39222.8 L Cabin 3: Hot water usage = 30778.5 L Cabin 4: Hot water usage = 16164.9 L The cabin with the highest hot water usage is Cabin 1, with a total of 45386.4 liters. Operationally, this suggests that Cabin 1 may have a higher demand for hot water compared to the other cabins. This could be due to factors such as more frequent use, longer showers, or higher temperature settings. It might be beneficial for the gym to monitor and optimize hot water usage in Cabin 1 to reduce energy costs and ensure an efficient water supply system.

Qwen2.5 3B (Local) · 48.4s

To compare hot vs cold water usage across all 4 gym shower cabins and calculate the hot-to-cold ratio per cabin: Cabin 1: Hot: 45,386.4 L Cold: 10,664.1 L Ratio: $45,386.4 / 10,664.1 = 4.27\times$ Cabin 2: Hot: 39,222.8 L Cold: 7,039.8 L Ratio: $39,222.8 / 7,039.8 = 5.56\times$ Cabin 3: Hot: 30,778.5 L Cold: 7,729.5 L Ratio: $30,778.5 / 7,729.5 = 4.01\times$ Cabin 4: Hot: 16,164.9 L Cold: 5,693.5 L Ratio: $16,164.9 / 5,693.5 = 2.83\times$ Now identifying which cabin uses the most hot water overall: Cabin 1: 45,386.4 L Cabin 2: 39,222.8 L Cabin 3: 30,778.5 L Cabin 4: 16,164.9 L The cabin that uses the most hot water overall is Cabin 1 with a total of 45,386.4 litres. This suggests operationally that: - Cabin 1 requires more energy and resources to heat water compared to other cabins. - It may be necessary for the gym management to consider reducing hot water usage in this cabin or find ways to optimize its operations to conserve hot water. - The high ratio indicates potential inefficiencies, such as leaks or inefficient heating systems that could be addressed for cost savings and environmental impact reduction.

[S3] Medium

Identify and explain the top anomalies in the residential CustomerC dataset. CustomerC has 7 sensors: Flush 1/2, Sink 1/2/3, Tap 1/2. A normal flush is ~6 L and a normal sink use is ~1-2 L. For each anomaly listed, explain the likely cause and recommended action.

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 1.3s

Based on the provided data, I have identified the top anomalies in the residential CustomerC dataset. I will explain the likely cause and recommended action for each anomaly. 1. **Anomaly:** Residential Flush 1 with consumption_L = 149.61 L, 137.01 L, 146.86 L, 200.65 L, 219.33 L, and 92.81 L. **Likely Cause:** These high consumption values are significantly higher than the normal flush consumption of ~6 L. This could be due to a malfunctioning flush valve or a clogged drain line. **Recommended Action:** Inspect the flush valve and drain line to identify and fix any issues. Consider replacing the flush valve if necessary. 2. **Anomaly:** Residential Sink 3 with consumption_L = 14.53 L and 82.84 L. **Likely Cause:** These high consumption values are significantly higher than the normal sink use of ~1-2 L. This could be due to a leaky faucet or a malfunctioning water sensor. **Recommended Action:** Inspect the faucet and water sensor to identify and fix any issues. Consider replacing the faucet or water sensor if necessary. 3. **Anomaly:** Residential Flush 1 with consumption_L = 92.81 L. **Likely Cause:** This high consumption value is significantly higher than the normal flush consumption of ~6 L. This could be due to a partial clog in the drain line or a malfunctioning flush valve. **Recommended Action:** Inspect the drain line and flush valve to identify and fix any issues. Consider replacing the flush valve if necessary. 4. **Anomaly:** Re... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Top Anomalies in CustomerC (Residential) (All events are single tap or single flush sessions. Normal values: Flush $\approx 6 \text{ L}$, Sink $\approx 1-2 \text{ L}$. All listed z-scores are > 12 , far above the typical ± 3 threshold.) * | # | Sensor | Date & Time | Consumption (L) | z-score | Likely Cause | Recommended Action | |---|---|---|---|---|---|---|---| | 1 | Flush 1 | 2024-06-25 13:13 | 149.61 | 14.3 | **Multiple flushes in one event** or a **sensor mis-trigger** (e.g., a long open tap that was counted as a single flush). | 1. Inspect the toilet for a stuck handle or a faulty flush valve. 2. Verify the sensor is correctly calibrated to detect a single flush. 3. If the sensor is fine, check for a leak in the toilet tank or bowl. | | 2 | Flush 1

Qwen3 32B (Groq) · 2.1s

<think> Okay, let's tackle this problem. The user wants me to identify and explain the top anomalies in the CustomerC dataset. They provided data from seven sensors: Flush 1/2, Sink 1/2/3, and Tap 1/2. The normal flush is around 6 liters, and a normal sink use is 1-2 liters. First, I need to look at the data given. The table has entries with high consumption_L values. The z-scores are all over 13, which is way higher than normal. Since a normal flush is 6L, the anomalies here are way beyond that. Looking at the first entry: Residential Flush 1 on 2024-06-25 with 149.61L. That's way more than 6L. The z-score is 14.3, which is extremely high. The same device appears multiple times with similar high consumption. The likely cause here could be a toilet leak or a faulty valve that's causing continuous water flow. The recommended action would be to check the toilet's fill valve or flapper. Next, the Sink 3 entries. On 2025-09-09, Sink 3 used 14.53L, and another entry on 2024-08-30 used 82.84L. Normal sink use is 1-2L, so these are anomalies. High consumption in a sink might be due to a dripping faucet or a stuck valve. The user should inspect the sink's taps for leaks or malfunctions. The highest consumption is 219.33L on 2024-06-27 for Flush 1. That's almost 36 times the normal amount. This could be a major leak in the plumbing system connected to the toilet. Immediate action is needed, like shutting off the water supply and checking for pipe bursts or faulty connections. T... [truncated]

Llama 3.3 70B (Groq) · 1.2s

Based on the provided data, the top anomalies in the residential CustomerC dataset are: 1. ****2024-06-25 13:13: Residential Flush 1 with 149.61 L****: This anomaly has a z-score of 14.3, indicating a significant deviation from the normal flush volume of ~6 L. The likely cause is a leaky flapper or a clogged drain line, causing excessive water flow. Recommended action: Inspect and repair the toilet flapper and drain line. 2. ****2024-05-22 09:41: Residential Flush 1 with 200.65 L****: This anomaly has a z-score of 13.36, indicating another significant deviation from the normal flush volume. The likely cause is similar to the previous anomaly, a leaky flapper or a clogged drain line. Recommended action: Inspect and repair the toilet flapper and drain line. 3. ****2024-06-27 20:32: Residential Flush 1 with 219.33 L****: This anomaly has a z-score of 12.86, indicating a significant deviation from the normal flush volume. The likely cause is again a leaky flapper or a clogged drain line. Recommended action: Inspect and repair the toilet flapper and drain line. 4. ****2024-08-30 08:24: Residential Sink 3 with 82.84 L****: This anomaly has a z-score of 13.33, indicating a significant deviation from the normal sink use volume of ~1-2 L. The likely cause is a faucet leak or an open faucet that was not turned off properly. Recommended action: Inspect and repair the faucet, and ensure it is turned off properly after use. 5. ****2025-09-09 13:36: Residential Sink 3 with 14.53 L****: This anomaly has a... [truncated]

Mistral 7B (Local) · 145.5s

In the provided CustomerC dataset, several anomalies can be identified based on the high z_score values, which indicate significant deviations from the rolling mean consumption for each device. Here are the top anomalies and their likely causes and recommended actions: 1. ****Residential Flush 1 (high consumption events)**** - Data time: 2024-06-25 13:13, 2024-07-10 08:10, 2024-06-27 20:32, 2024-08-30 08:24, 2025-09-09 13:36, and 2024-25-02-22 09:41 - Consumption_L: 149.61 L, 92.81 L, 219.33 L, 82.84 L, 200.65 L, and 14.53 L (respectively) - Rolling mean: 5.xx L (varies for each event) - z_score: 12.86 to 14.3 (extremely high values) These events show significantly higher-than-normal water consumption during flushes, which could be due to factors such as a faulty toilet, a long flush duration, or multiple flushes in quick succession. Recommended actions include: - Investigating the toilet for any mechanical issues (e.g., worn-out parts, improperly adjusted float valve) - Monitoring the flush duration and considering implementing a flush limit to reduce excessive water usage - Educating the customer about proper flushing techniques to minimize water waste 2. ****Residential Sink 3 (high consumption event)**** - Data time: 2024-10-08 08:52 and 2024-09-08 13:36 - Consumption_L: 137.01 L and 14.53 L (respectively) - Rolling mean: 0.63 L and 0.63 L (respectively) - z_score: 13.62 and 13.33 (extremely high values) These events show significantly... [truncated]

Qwen2.5 3B (Local) · 83.0s

Let's analyze the anomalies in the provided data for CustomerC: **### Anomaly 1: **Data Time:** 2024-06-27 20:32 **Device Label:** Residential Flush 1 **Consumption_L:** 219.33 L **Roll Mean:** 5.63 L/min (normal flush is ~6 L) **Z-Score:** -12.86 **Likely Cause:** This anomaly suggests an extremely high consumption rate for a Flush event, which is not typical and could be due to a malfunction or misuse of the sensor. **Recommended Action:** Investigate this event further by checking if there was any unusual activity at that time (e.g., someone tampering with the sensor). If confirmed as a misreport, consider replacing the sensor. Otherwise, it might indicate an issue with the plumbing system and should be reported to maintenance for inspection. **### Anomaly 2: **Data Time:** 2024-05-22 09:41 **Device Label:** Residential Flush 1 **Consumption_L:** 200.65 L **Roll Mean:** 6.62 L/min (normal flush is ~6 L) **Z-Score:** -13.36 **Likely Cause:** This anomaly also indicates an extremely high consumption rate for a Flush event, which could be due to the same reasons as Anomaly 1. **Recommended Action:** Similar to Anomaly 1, investigate this event further by checking if there was any unusual activity at that time (e.g., someone tampering with the sensor). If confirmed as a misreport, consider replacing the sensor. Otherwise, it might indicate an issue with the plumbing system and should be reported to maintenance for inspection. **### Anomaly 3: **Data Time:** ... [truncated]******

[S4] Medium

Which customer has the highest average daily water consumption? Rank all 4 customers and explain the difference given their profiles: Gym (showers), CustomerA (office toilet bloc), CustomerB (sanitary bloc with wudu sink), CustomerC (residential home).

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.7s

To find the customer with the highest average daily water consumption, we need to rank the customers based on their average daily consumption. 1. CustomerB: 1003.4 L/day 2. CustomerA: 604.1 L/day 3. Gym: 218.1 L/day 4. CustomerC: 258.1 L/day However, CustomerC has a higher average daily consumption than the Gym. So, the correct ranking is: 1. CustomerB: 1003.4 L/day 2. CustomerA: 604.1 L/day 3. CustomerC: 258.1 L/day 4. Gym: 218.1 L/day The difference between CustomerB and CustomerA is 399.3 L/day. This is likely due to the fact that CustomerB has a sanitary bloc with a wudu sink, which requires more water for cleaning and flushing. The difference between CustomerC and the Gym is 40 L/day. This is relatively small, considering the different types of usage. CustomerC is a residential home, which typically requires more water for various household activities. The Gym, on the other hand, is likely to have showers, but the usage is relatively low compared to CustomerC. The difference between CustomerC and CustomerA is 346 L/day. This is likely due to the fact that CustomerA has an office toilet bloc, which requires more water for flushing and cleaning compared to a residential home.

GPT-OSS 20B (Groq) · 0.8s

****Ranking of customers by average daily water consumption**** | Rank | Customer | Avg. daily consumption (L/day) |
 |-----|-----|-----| 1 | ****Customer B**** | 1003.4 | 2 | ****Customer A**** | 604.1 | 3 | ****Customer C**** | 258.1 | 4 |
****Gym**** | 218.1 | ****Why the differences?**** | Customer | Profile | What drives the numbers | |-----|-----|-----|
****Customer B**** | Sanitary bloc with a wudu sink | The daily average is the highest. Even though a wudu sink is a small fixture, the block likely includes other high-flow uses (e.g., handwashing stations, small showers, or frequent cleaning). The large total volume (214,731.3 L over 214 days) shows that the block is used heavily, possibly by many people or for repeated sanitation tasks. |
****Customer A**** | Office toilet bloc | The average of 604.1 L/day is high for a toilet bloc, reflecting a busy office environment with many employees. The total volume (105,712.5 L over 175 days) indicates consistent daily use, but lower than Customer B because the office may have fewer high-flow fixtures than the sanitary block. | ****Customer C**** | Residential home | A typical home uses water mainly for showers, dishwashing, and laundry. The average of 258.1 L/day is moderate, reflecting normal household habits. The long observation period (628 days) gives a stable average. | ****Gym**** | Showers | Although a gym has showers, the average of 218.1 L/day is the lowest. This could be due to... [truncated]

Qwen3 32B (Groq) · 2.0s

<think> Okay, let's tackle this problem step by step. The user wants to know which customer has the highest average daily water consumption and then rank all four customers. The profiles given are Gym (showers), CustomerA (office toilet bloc), CustomerB (sanitary bloc with wudu sink), and CustomerC (residential home). First, I need to look at the data provided. The data includes the average daily consumption for each customer and the total consumption over a certain number of days. The key here is that the average daily consumption is already provided, so I don't need to calculate it. The question is straightforward: which of these averages is the highest? Looking at the numbers: - CustomerA: 604.1 L/day - CustomerB: 1003.4 L/day - CustomerC: 258.1 L/day - Gym: 218.1 L/day From these, CustomerB has the highest average daily consumption at 1003.4 L/day. The next step is to rank all four customers from highest to lowest. So the order would be CustomerB, CustomerA, CustomerC, and then Gym. Now, the user also wants an explanation of the differences based on their profiles. Let's think about each customer's usage scenario. CustomerB is a sanitary bloc with a wudu sink. Wudu sinks are used for ritual washing in Islamic practices, which might involve more frequent and possibly longer water use compared to regular sinks. This could explain the high daily consumption. CustomerA is an office toilet bloc. Office settings typically have a high number of users, but the water usage... [truncated]

Llama 3.3 70B (Groq) · 0.9s

To determine the customer with the highest average daily water consumption, we need to look at the provided data. The average daily water consumption for each customer is already given: - customerA: 604.1 L/day - customerB: 1003.4 L/day - customerC: 258.1 L/day - gym: 218.1 L/day Ranking the customers from highest to lowest average daily water consumption: 1. customerB: 1003.4 L/day 2. customerA: 604.1 L/day 3. customerC: 258.1 L/day 4. gym: 218.1 L/day The difference in water consumption can be explained by the profiles of each customer: - customerB has a sanitary bloc with a wudu sink, which likely requires more water for cleaning and hygiene, resulting in the highest average daily consumption. - customerA has an office toilet bloc, which would have a moderate number of users and flushing events, resulting in the second-highest average daily consumption. - customerC is a residential home, which typically has lower water usage due to fewer occupants and more conservative water use habits, resulting in a lower average daily consumption. - The gym has showers, but its average daily consumption is the lowest, possibly due to the showers being used intermittently and for shorter periods, or the gym having water-efficient showerheads.

Mistral 7B (Local) · 167.6s

To find the average daily water consumption for each customer, we first need to calculate their total session durations and then divide their total litres used by the total session durations in seconds. After that, we can rank them based on their average daily consumption. 1. CustomerA: Total litres used = 105,712.5 L Total session duration (in seconds) = Not provided in the data However, since there are 175 days of data, we can assume that the customer uses water every day. So, let's calculate the average number of usage events per day: Average events per day = Total days / Total days with data = 365 / 175 = 2.05 events/day Now, let's assume that each event lasts for an average of 60 seconds (this is just an assumption since the data doesn't provide session durations): Total session duration (in

seconds) = Average events per day * Average event duration = 2.05 * 60 ≈ 123 seconds/day Finally, let's calculate the average daily water consumption: Average daily water consumption = Total litres used / Total session duration (in seconds) ≈ 105,712.5 L / (123 * 175) ≈ 604.1 L/day 2. CustomerB: Total litres used = 214,731.3 L Total session duration (in seconds) = Not provided in the data Let's assume an average of 2 events per day and each event lasts for an average of 60 seconds: Total session duration (in seconds) = Average events per day * Average event duration = 2 * 60 ≈ 120 seconds/day Average daily water consumption = Total litres used... [truncated]

Qwen2.5 3B (Local) · 64.1s

To determine which customer has the highest average daily water consumption, we need to calculate the average daily usage for each of the four customers based on their total monthly and total annual water usage. ### CustomerA (Office Toilet Bloc) - Total Water Usage: 105,712.5 L over 175 days - Average Daily Consumption: $-\left(\frac{105,712.5}{175}\right) = 603.54$ L/day ### CustomerB (Sanitary Bloc with Wudu Sink) - Total Water Usage: 214,731.3 L over 214 days - Average Daily Consumption: $-\left(\frac{214,731.3}{214}\right) = 1003.4$ L/day ### CustomerC (Residential Home) - Total Water Usage: 162,082.1 L over 628 days - Average Daily Consumption: $-\left(\frac{162,082.1}{628}\right) = 259.74$ L/day ### Gym (Showers) - Total Water Usage: 162,679.6 L over 746 days - Average Daily Consumption: $-\left(\frac{162,679.6}{746}\right) = 218.05$ L/day ### Ranking of Customers by Average Daily Water Usage: 1. **CustomerB (Sanitary Bloc with Wudu Sink)**: - Highest average daily consumption at 1003.4 L/day. 2. **CustomerC (Residential Home)**: - Second highest, averaging 259.74 L/day. 3. **Gym (Showers)**: - Third with an average of 218.05 L/day. 4. **CustomerA (Office Toilet Bloc)**: - Lowest at 603.54 L/day. ### Explanation of Differences: **CustomerB** and **CustomerC** both have high water usage, but CustomerB uses significantly more per day due to the higher total volume over a shorter period compared to CustomerC. - **CustomerA** has an average daily consumption of 603.54... [truncated]

[S5] Medium

Compare CustomerA and CustomerB water consumption side by side. CustomerA is an office toilet bloc with 4 aggregated bloc sensors. CustomerB is a sanitary bloc (2 WCs, sinks, wudu sink) with 1 aggregated sensor. Explain the difference in total and daily consumption. Which is more water-efficient per...

✓ Real sensor data injected into this prompt

Llama 3.1 8B (Groq) · 0.7s

To compare the water consumption of CustomerA and CustomerB, we need to calculate the total consumption and daily consumption for each customer. **Total Consumption:** CustomerA: 105,712.5 L CustomerB: 214,731.3 L CustomerB has a higher total consumption of 109,018.8 L (214,731.3 L - 105,712.5 L). **Daily Consumption:** To calculate the daily consumption, we need to know the duration of the data collection period for each customer. CustomerA: 2025-05-31 → 2025-11-22 = 176 days (approximate) CustomerB: 2025-10-06 → 2026-05-09 = 220 days (approximate) CustomerA's daily consumption = 105,712.5 L / 176 days ≈ 601.1 L/day CustomerB's daily consumption = 214,731.3 L / 220 days ≈ 976.3 L/day CustomerB has a higher daily consumption of 375.2 L/day (976.3 L/day - 601.1 L/day). **Water Efficiency per User:** To compare the water efficiency per user, we need to know the number of users for each customer. Since we don't have this information, we'll assume that the number of users is proportional to the number of sensors. CustomerA has 4 sensors, and CustomerB has 1 sensor. This suggests that CustomerA has 4 times more users than CustomerB. However, this is not a direct comparison, as the water usage patterns may differ between the two customers. To make a fair comparison, we would need more information about the number of users, usage patterns, and other factors. Based on the available data, we can conclude that CustomerB has a higher total and daily consumption than CustomerA... [truncated]

GPT-OSS 20B (Groq) · 1.0s

Side by side comparison | Metric | Customer A (office toilet bloc) | Customer B (sanitary bloc) |
|-----|-----|-----| | Total consumption | 105,712.5 L | 214,731.3 L | | Period covered |
2025-05-31 → 2025-11-22 | 2025-10-06 → 2026-05-09 | | Days in period | 176 days | 216 days | | Daily average | 105,712.5 L ÷ 176 days = 600.4 L / day | 214,731.3 L ÷ 216 days = 994 L / day | | Sensors (users) | 4 | 1 | | Consumption per sensor | 105,712.5 L ÷ 4 = 26,428.1 L | 214,731.3 L ÷ 1 = 214,731.3 L | | Daily consumption per sensor | 600

Qwen3 32B (Groq) · 1.5s

<think> Okay, let's tackle this problem step by step. First, I need to compare the water consumption of CustomerA and CustomerB. The user wants to know the difference in total and daily consumption and determine which is more water-efficient per user. Starting with the data provided: CustomerA used 105,712.5 liters from May 31, 2025, to November 22, 2025, with 4 sensors. CustomerB used 214,731.3 liters from October 6, 2025, to May 9, 2026, with 1 sensor. First, I should calculate the total consumption for each. CustomerA's total is given as 105,712.5 L, and CustomerB's is 214,731.3 L. That's straightforward. The next step is to find the daily consumption. To do that, I need the number of days each customer was monitored. For CustomerA, the period is from May 31 to November 22. Let me count the days. May has 31 days, so from May 31 to May 31 is 1 day. Then June has 30, July 31, August 31, September 30, October 31, and November 22. Adding those up: 1 + 30 + 31 + 31 + 30 + 31 + 22. Let me calculate that: 1+30=31, +31=62, +31=93, +30=123, +31=154, +22=176 days. Wait, but May 31 to November 22 is actually from May 31 to November 22 inclusive? Wait, no. If the start date is May 31 and the end date is November 22, the total days would be from May 31 to November 22. Let me check again. May 31 is day 1, then June has 30 days, July 31, August 31, September 30, October 31, November 22. So total days: 30 (June) + 31 (July) + 31 (August) + 30 (September) + 31 (October) + 22 (November)... [truncated]

Llama 3.3 70B (Groq) · 0.7s

To compare CustomerA and CustomerB water consumption, we need to calculate the total days of data and daily consumption for each customer. CustomerA data period: 2025-05-31 → 2025-11-22 = 205 days (from May 31 to November 22) CustomerA total

consumption: 105,712.5 L CustomerA daily consumption: 105,712.5 L / 205 days = 515.07 L/day CustomerB data period: 2025-10-06 → 2026-05-09 = 216 days (from October 6 to May 9) CustomerB total consumption: 214,731.3 L CustomerB daily consumption: 214,731.3 L / 216 days = 994.34 L/day CustomerA has a lower total consumption (105,712.5 L) and daily consumption (515.07 L/day) compared to CustomerB (214,731.3 L and 994.34 L/day). To determine which is more water-efficient per user, we need more information about the number of users for each customer. However, based on the available data, CustomerA appears to be more water-efficient overall, given its lower daily consumption.

Mistral 7B (Local) · 133.7s

To compare the water consumption of CustomerA and CustomerB, let's first calculate their total water usage and then their daily average usage. Customer A's data covers a longer period (from May 31, 2025 to November 22, 2025) compared to Customer B's data (from October 6, 2025 to May 9, 2026). To make the comparison fair, let's focus on the overlapping period between both datasets, which is from October 6, 2025 to November 22, 2025. During this time, Customer A used a total of 105,712.5 L of water. Since there are 4 sensors for CustomerA, the average consumption per sensor can be calculated as follows: $\text{average_consumption_A} = \text{total_consumption_A} / \text{number_of_sensors_A}$ $\text{average_consumption_A} = 105712.5 \text{ L} / 4 = 26428.13 \text{ L}$ per sensor Customer B used a total of 214,731.3 L of water during the same period (from October 6, 2025 to November 22, 2025). Since there is only 1 sensor for CustomerB, the total consumption can be considered as the average consumption per user: $\text{average_consumption_B} = \text{total_consumption_B}$ $\text{average_consumption_B} = 214731.3 \text{ L}$ Now let's calculate the daily average consumption for both customers during the overlapping period. First, we need to find out how many days are covered in this period: $\text{days_overlapped} = \text{number_of_days_between}(\text{November 22, 2025}) \text{ and } (\text{October 6, 2025}) + 1$ $\text{days_overlapped} = 84$ days Next, we can calculate the daily average consumption for both customers: $\text{daily_average_consumption_A} = \text{average_consumption_A} / \text{days_overlapped}$ $\text{daily_average_consumption_A} = 26428.13 \text{ L} / 84 \text{ days} \approx 314.62 \text{ L/day}$ [truncated]

Qwen2.5 3B (Local) · 59.4s

To compare the water consumption and efficiency of CustomerA (office toilet bloc) and CustomerB (sanitary bloc), we need to calculate their total and daily average water usage, as well as determine which is more water-efficient per user. #### Total Water Usage **CustomerA**: - Total consumption: 105,712.5 L - Duration of data collection: 489 days (from May 31 to November 22) **CustomerB**: - Total consumption: 214,731.3 L - Duration of data collection: 665 days (from October 6 to May 9) #### Daily Average Water Usage **CustomerA**: - Daily average = 105,712.5 L / 489 days $\approx 216.4 \text{ L/day}$ **CustomerB**: - Daily average = 214,731.3 L / 665 days $\approx 324.9 \text{ L/day}$ #### Water Efficiency per User Assuming each sensor represents one user (office toilet or sanitary facility): **CustomerA**: - Number of users: 4 - Average water usage per user = 105,712.5 L / 4 $\approx 26,428.13 \text{ L/user}$ **CustomerB**: - Number of users: 1 (sanitary bloc) - Average water usage per user = 214,731.3 L / 1 $\approx 214,731.3 \text{ L/user}$ #### Comparison and Analysis **Total Water Usage**: - CustomerA used approximately 105,712.5 L - CustomerB used approximately 214,731.3 L CustomerB uses significantly more water (214,731.3 L vs 105,712.5 L). **Daily Average Water Usage**: - CustomerA's daily average is about 216.4 L/user/day - CustomerB's daily average is about 324.9 L/user/day CustomerB uses more water per day (324.9 L/user/day vs 216.4 L/user/day). **Water Efficiency**: - CustomerA's average usage per user: $\approx 26,428.13 \text{ L}$ [truncated]

Tool-Use Prompts

[T1] Easy

Compare hot vs cold water usage across all gym shower cabins. To answer this you must: 1. Call `query_data` with `customer='gym'`, `sub_category='Hot'` to get hot totals 2. Call `query_data` with `customer='gym'`, `sub_category='Cold'` to get cold totals 3. Call `generate_chart` with `chart_type='compare'`, `customer='gym'`

Llama 3.1 8B (Groq) · 0.3s

To compare hot vs cold water usage across all gym shower cabins, I would make the following tool calls: 1. `query_data`: - Tool call: `query_data(customer='gym', sub_category='Hot')` - This will return the total litres used for hot water in the gym shower cabins. 2. `query_data`: - Tool call: `query_data(customer='gym', sub_category='Cold')` - This will return the total litres used for cold water in the gym shower cabins. 3. `generate_chart`: - Tool call: `generate_chart(chart_type='compare', customer='gym')` - This will generate a chart comparing the hot and cold water usage in the gym shower cabins. These tool calls will provide the necessary data to compare the hot and cold water usage across all gym shower cabins.

GPT-OSS 20B (Groq) · 0.4s

I would make the following tool calls: 1. `query_data` *Arguments*: - `customer = 'gym'` - `sub_category = 'Hot'` 2. `query_data` *Arguments*: - `customer = 'gym'` - `sub_category = 'Cold'` 3. `generate_chart` *Arguments*: - `chart_type = 'compare'` - `customer = 'gym'`

Qwen3 32B (Groq) · 0.9s

<think> Okay, let's tackle this problem step by step. The user wants to compare hot and cold water usage across all gym shower cabins. First, I need to figure out how to get the necessary data. The instructions say to call `query_data` twice: once for 'Hot' and once for 'Cold' under the 'gym' customer. That makes sense because we need separate totals for each. So, the first step is to make those two queries. After getting the data, the next step is to generate a comparison chart using `generate_chart` with the 'compare' type and the 'gym' customer. I should make sure the arguments are correctly specified for each function call. Wait, do I need to process the data in any way before generating the chart? The user just wants the totals, so maybe the `query_data` functions will return aggregated numbers. The `generate_chart` function probably takes those totals and creates a visual comparison. I need to list each tool call with their respective arguments. Let me check the parameters again. For `query_data`, the parameters are `customer` and `sub_category`. For `generate_chart`, it's `chart_type` and `customer`. All right, that's straightforward. No need to do any calculations here since the user hasn't

provided actual data yet. Just the structure of the tool calls is required. So the answer should be three tool calls: two queries and one chart generation. I should present each call in order, making sure the parameters are correctly assigned. Let me write them out clearly.

</think> 1. que... [truncated]

Llama 3.3 70B (Groq) · 0.3s

To compare hot vs cold water usage across all gym shower cabins, I would make the following tool calls: 1. query_data with the following arguments: customer = 'gym', sub_category = 'Hot' 2. query_data with the following arguments: customer = 'gym', sub_category = 'Cold' 3. generate_chart with the following arguments: chart_type = 'compare', customer = 'gym'

Mistral 7B (Local) · 42.5s

To compare hot vs cold water usage across all gym shower cabins, I would follow these steps: 1. First, I would call the 'query_data' function twice with the following arguments for each call: - For hot water usage: 'customer='gym'', 'sub_category='Hot''. This will return a list of total litres used ('consumption_L') and session durations ('session_duration_s') for all gym shower cabins using hot water. - For cold water usage: 'customer='gym'', 'sub_category='Cold''. This will return a similar list of total litres used and session durations for all gym shower cabins using cold water. 2. After obtaining the data, I would calculate the average consumption (in litres) and average session duration (in seconds) for both hot and cold water usage separately. 3. Finally, I would call the 'generate_chart' function with the following arguments: - 'chart_type='compare'' to create a comparison chart - 'customer='gym'' to specify that the data is for gym shower cabins. By following these steps, I will be able to compare hot vs cold water usage across all gym shower cabins and visualize the results using a comparison chart.

Qwen2.5 3B (Local) · 18.8s

To compare hot vs cold water usage across all gym shower cabins, I would need to call two functions: 'query_data' for retrieving totals and 'generate_chart' for visualizing the data. Here are the steps with the corresponding function calls: 1. **Call query_data with customer='gym', sub_category='Hot': - Arguments: - customer: 'gym' - sub_category: 'Hot' - This will fetch the total litres used for hot water in all gym shower cabins. 2. **Call query_data with customer='gym', sub_category='Cold': - Arguments: - customer: 'gym' - sub_category: 'Cold' - This will fetch the total litres used for cold water in all gym shower cabins. 3. **Call generate_chart with chart_type='compare', customer='gym': - Arguments: - chart_type: 'compare' - customer: 'gym' - This will create a visual comparison chart showing hot and cold water usage across all gym shower cabins, based on the data fetched from the previous calls. These function calls are designed to provide the necessary information for comparing hot vs cold water usage in the gym's shower cabins.

[T2] Medium

Plot the daily consumption trend for customerC over the past 3 months. To answer this you must: 1. Call query_data with customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24' 2. Call generate_chart with chart_type='line', customer='customerC', group_by='day', start_date=...

Llama 3.1 8B (Groq) · 0.6s

To plot the daily consumption trend for customerC over the past 3 months, I would use the following tool calls: 1. Query Data: ```python query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') ``` This will retrieve the daily consumption data for customerC from September 1st to November 24th. 2. Generate Chart: ```python generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') ``` This will generate a line chart to display the daily consumption trend for customerC over the specified period. Note: The 'generate_chart' function will use the data retrieved by the 'query_data' function to create the chart. Assuming the data is in the following format: | date | consumption_L | |-----|-----| 2025-09-01 | 100.0 | 2025-09-02 | 120.0 | ... | ... | 2025-11-24 | 150.0 | The chart will display the daily consumption trend for customerC over the past 3 months.

GPT-OSS 20B (Groq) · 0.4s

```
query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')
generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')
```

Qwen3 32B (Groq) · 1.4s

<think> Okay, let's see. The user wants me to plot the daily consumption trend for customerC over the past three months. The data should be from September 1, 2025, to November 24, 2025. First, I need to call the query_data function. The parameters required are customer, group_by, start_date, and end_date. The customer is customerC, group_by is day, and the dates are as specified. So the first tool call would be query_data with those arguments. Next, after getting the data, I need to generate a line chart using generate_chart. The chart_type is line, and the same customer and dates apply. The group_by is still day. So the second call is generate_chart with those parameters. I need to make sure the dates are correct. The user mentioned that customerC's data ends on 2025-11-24, so the end_date is fixed. The start_date is three months prior, which would be September 1, 2025. That makes sense because September to November is three months. I should check if there are any other parameters needed, but the problem statement doesn't mention any. The key is to use the exact dates and customer ID provided. No need to adjust the dates relative to today since the end_date is given as 2025-11-24. So the two tool calls are straightforward. Just pass the correct parameters each time. No calculations are needed here, just formatting the function calls correctly. </think> 1. query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') 2. generate_cha... [truncated]

Llama 3.3 70B (Groq) · 0.5s

To plot the daily consumption trend for customerC over the past 3 months, I would use the following tool calls and arguments: 1. query_data call: query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') This call will retrieve the daily consumption data for customerC from 2025-09-01 to 2025-11-24. 2. generate_chart call:

generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24') This call will generate a line chart to visualize the daily consumption trend for customerC over the specified period.

Mistral 7B (Local) · 45.0s

To plot the daily consumption trend for customerC over the past 3 months (from September 1, 2025 to November 24, 2025), I would use the following code: 1. Call query_data with the following arguments: `query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')` This call will return a dataframe containing the total litres consumed (consumption_L) and session duration (session_duration_s) for each day within the specified date range. 2. Call generate_chart with the following arguments: `generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')` This call will create a line chart displaying the daily consumption trend for customerC over the specified date range. The x-axis will represent the days, and the y-axis will represent the litres consumed per day.

Qwen2.5 3B (Local) · 27.4s

To achieve the task of plotting the daily consumption trend for customer 'customerC' over the past 3 months, we need to follow these steps: **Step 1: Query Data** We will use the `query_data` tool with the following arguments: - `customer`: Set to `'customerC'` - `group_by`: Set to `'day'` - `start_date`: Set to `'2025-09-01'` - `end_date`: Set to `'2025-11-24'` **Tool Call:** `query_data(customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')` **Step 2: Generate Chart** After obtaining the data from the `query_data` tool, we will use the `generate_chart` tool with these arguments: - `chart_type`: Set to `'line'` - `customer`: Set to `'customerC'` - `group_by`: Set to `'day'` - `start_date`: Set to `'2025-09-01'` - `end_date`: Set to `'2025-11-24'` **Tool Call:** `generate_chart(chart_type='line', customer='customerC', group_by='day', start_date='2025-09-01', end_date='2025-11-24')` These tool calls will provide the necessary data and visualization for the daily consumption trend of customer 'customerC' over the specified period.

[T3] Hard

Detect anomalies for customerC and then visualise the result as a line chart. To answer this you must: 1. Call detect_anomalies with customer='customerC', window_days=7 2. Call generate_chart with chart_type='line', customer='customerC' to show the consumption trend with context IMPORTANT: Never pas...

Llama 3.1 8B (Groq) · 1.2s

To detect anomalies and generate a line chart, I'll use the following tool calls: 1. `detect_anomalies(customer='customerC', window_days=7, device_label='all')` **WRONG** - omit `device_label='all'` to include all devices. Corrected call: `detect_anomalies(customer='customerC', window_days=7)` 2. `generate_chart(chart_type='line', customer='customerC', device_label='all')` **WRONG** - omit `device_label='all'` to include all devices. Corrected call: `generate_chart(chart_type='line', customer='customerC')` Here are the Python code snippets for the above calls: `python # Assuming 'detect_anomalies' and 'generate_chart' are functions anomalies = detect_anomalies(customer='customerC', window_days=7) print(anomalies) # Generate line chart for customerC generate_chart(chart_type='line', customer='customerC')` Please note that I'm assuming `'detect_anomalies'` and `'generate_chart'` are functions that are already defined and available for use. If not, you'll need to define them or import the necessary libraries. Here's a sample implementation of the `'detect_anomalies'` function: `python def detect_anomalies(customer, window_days): # Assuming 'data' is a list of dictionaries containing consumption data data = [{'customer': 'customerA', 'device': 'device1', 'consumption_L': 100, 'session_duration_s': 3600}, {'customer': 'customerA', 'device': 'device1', 'consumption_L': 120, 'session_duration_s': 4200}, {'customer': 'customerC', 'device': 'device1', 'consumption_L': 100, 'session_duration_s': 3600}, {'customer': 'customerC', 'device': 'device1', 'consumption_L': 120, 'session_duration_s': 4200}, ...]`

GPT-OSS 20B (Groq) · 0.5s

Tool calls 1. `detect_anomalies(customer='customerC', window_days=7)` 2. `generate_chart(chart_type='line', customer='customerC')` **Note:** The `device_label` argument is omitted entirely, as passing `device_label='all'` is disallowed.

Qwen3 32B (Groq) · 1.7s

`<think>` Okay, let's tackle this problem step by step. The user wants me to detect anomalies for customerC and then visualize the result as a line chart. The instructions specify using the `detect_anomalies` function with `customer='customerC'` and `window_days=7`, and then `generate_chart` with `chart_type='line'` and `customer='customerC'`. They also mention not to pass `device_label='all'` and to flag any wrong arguments. First, I need to make sure I call `detect_anomalies` correctly. The parameters are `customer='customerC'` and `window_days=7`. The `device_label` should be omitted, so I won't include it. That's correct because the user said to omit `device_label` to include all devices. So the first function call is `detect_anomalies(customer='customerC', window_days=7)`. Next, the `generate_chart` function needs to be called with `chart_type='line'` and `customer='customerC'`. Again, `device_label` should not be passed as `'all'` or anything else. The user emphasized that `device_label='all'` is wrong, so I must exclude it. Therefore, the second call is `generate_chart(chart_type='line', customer='customerC')`. I should check if there are any other parameters that might be incorrect. The user didn't mention other parameters, so assuming the defaults are okay. Also, the `chart_type` is specified as `'line'`, which is correct for a line chart. The customer is correctly set to `'customerC'`. Wait, the user said to "flag any argument that would be WRONG to pass." In this case, if someone were to include `device_label='all'`...

Llama 3.3 70B (Groq) · 0.6s

To detect anomalies and visualize the result for customerC, I will make the following tool calls: 1. `detect_anomalies(customer='customerC', window_days=7)` 2. `generate_chart(chart_type='line', customer='customerC')` I will not pass the argument `device_label='all'` as it is incorrect to pass this argument when including all devices. Omitting the `device_label` argument entirely will include all devices. The exact tool calls are: - `detect_anomalies(customer='customerC', window_days=7)` - `generate_chart(chart_type='line', customer='customerC')` Passing `device_label='all'` would be WRONG, as it should be omitted to include all devices.

Mistral 7B (Local) · 32.2s

To detect anomalies for customer 'customerC' and visualize the result as a line chart, follow these steps: 1. Call `detect\_anomalies` with the following arguments: - customer: 'customerC' - window_days: 7 Here is how it would look: ```python detect_anomalies(customer='customerC', window_days=7) ``` 2. Call `generate\_chart` with the following arguments: - chart_type: 'line' - customer: 'customerC' Since we don't need to include all devices, we omit the device_label argument entirely: ```python generate_chart(chart_type='line', customer='customerC') ```

Qwen2.5 3B (Local) · 16.4s

To detect anomalies for customer 'customerC' over a 7-day window and visualize the result as a line chart, I will use the following commands: 1. ****detect_anomalies**** with `customer='customerC', window\_days=7`: - Correct call: `detect\_anomalies(customer='customerC', window\_days=7)` - Incorrect calls (do not pass these): - `detect\_anomalies(window\_days=7)` // Missing customer parameter - `detect\_anomalies(customer='all', window\_days=7)` // Passing device_label='all' is wrong 2. ****generate_chart**** with `chart\_type='line', customer='customerC': - Correct call: `generate_chart(chart_type='line', customer='customerC')` - Incorrect calls (do not pass these): - `generate_chart(customer='all', chart_type='line')` // Passing device_label='all' is wrong Please execute the above commands to achieve the desired outcome.

5 Summary Scorecard

Model	Avg Score	Avg Time	Errors	Monthly \$
Llama 3.1 8B (Groq)	0.000	0.9s	0	\$0.15
GPT-OSS 20B (Groq)	0.000	0.9s	0	\$0.66
Qwen3 32B (Groq)	0.000	1.9s	0	\$1.01
Llama 3.3 70B (Groq)	0.000	1.2s	0	\$1.63
Mistral 7B (Local)	0.000	102.0s	0	\$0.00
Qwen2.5 3B (Local)	0.000	58.5s	0	\$0.00

6 Cost–Quality Recommendation

Budget score = quality ÷ (1 + log(1 + monthly cost)). Higher = better quality per dollar.

	Model	Avg Quality	Monthly \$	Budget Score
★	Llama 3.1 8B (Groq)	0.000	\$0.15	0.000
2	GPT-OSS 20B (Groq)	0.000	\$0.66	0.000
3	Qwen3 32B (Groq)	0.000	\$1.01	0.000
	Llama 3.3 70B (Groq)	0.000	\$1.63	0.000
	Mistral 7B (Local)	0.000	\$0.00	0.000
	Qwen2.5 3B (Local)	0.000	\$0.00	0.000

★ Recommendation: Llama 3.1 8B (Groq) offers the best quality-to-cost balance. For on-premises deployments, prefer the highest-scoring local model.

7 Proposed Production Fallback Chain

- 1 Llama 3.1 8B (Groq) — Best budget score
- 2 GPT-OSS 20B (Groq) — Rate-limit fallback
- 3 Qwen3 32B (Groq) — Rate-limit fallback
- 4 Llama 3.3 70B (Groq) — Rate-limit fallback
- 5 Mistral 7B (Local) — Local fallback — zero cost / data privacy
- 6 Qwen2.5 3B (Local) — Local fallback — zero cost / data privacy

Groq keys rotate automatically across KEY_1–KEY_4. Reorder _PROVIDERS in llm_client.py to change the agent default.